

The Inference is the Search: AI-Driven Predictive Surveillance and the Fourth Amendment

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**“ THE TRANSFORMATION OF
AGGREGATED PERSONAL DATA INTO A
PERSON-SPECIFIC RISK SCORE OR
IDENTITY MATCH REQUIRES NO LESS
THAN A WARRANT. ”**

Abstract

This article argues that when law enforcement uses an AI system to generate a person specific probabilistic inference and acts on that inference, it has conducted a search within the meaning of the Fourth Amendment requiring a warrant or established exception. Existing doctrine, from *Katz v. United States* through *United States v. Jones* to *Carpenter v. United States*, has moved progressively closer to this problem without resolving it. *Carpenter* held that the comprehensiveness of digital surveillance can trigger Fourth Amendment protection regardless of the third-party doctrine, but addressed only data collection, not the AI synthesis of collected data into actionable person-specific outputs. This article fills that gap. It develops four arguments: that the constitutional harm occurs when a person-specific inference is used as the basis for law enforcement action, not when the underlying data is collected; that AI inference is categorically different from human analysis in ways that make constitutional protection more urgent; that probabilistic outputs are structurally incompatible with the probable cause standard; and that AI systems' documented capacity to hallucinate with confidence makes judicial oversight more urgent, not less. It then proposes a narrow, judicially administrable standard requiring a warrant before law enforcement takes investigative action based on a person-specific AI output. The argument does not require courts to break new ground. It requires them to apply the logic of *Carpenter* to the next question *Carpenter* did not answer.

I. Introduction

In February 2019, Nijeer Parks was arrested for a crime he did not commit. Police identified him not through a witness or physical evidence, but through a facial recognition algorithm that matched his photo to a blurry surveillance image from a hotel gift shop in Woodbridge, New Jersey. The match was wrong. Parks had never owned a car and had never held a driver's license, yet he stood accused of shoplifting and fleeing the scene in a vehicle. He turned himself in to clear up the mistake and was arrested on the spot. He spent ten days in jail and paid approximately five thousand dollars to defend himself before the case was dismissed for lack of evidence. Parks is not alone. [1] Robert Williams in Detroit, Michael Oliver in New Orleans, and Porcha Woodruff, also in Detroit, were each wrongfully arrested after police acted on a facial recognition match without independent verification. In each case, the same constitutional question went unanswered: when law enforcement uses an AI-generated output to target a specific person and acts on that output, has it conducted a search under the Fourth Amendment?

The legal framework that should govern cases like Parks' remains undeveloped. The Fourth Amendment guarantees the right of people to be secure against unreasonable searches and seizures, and for decades courts have asked a single question when evaluating digital surveillance: was there a search when the data was collected? In *Carpenter v. United States*, the Supreme Court took a significant step forward, holding that police must obtain a warrant before accessing long-term cell phone location data and acknowledging that seismic shifts in digital technology may require rethinking foundational Fourth Amendment doctrine. [2] But *Carpenter* was deliberately narrow. The Court addressed data collection by third parties and explicitly declined to disturb conventional surveillance techniques or address other collection technologies. It said nothing about what happens after collection, when an AI system ingests that data and synthesizes it into a probabilistic inference targeting a specific individual. That inference, the output the government acts on, the score or match that puts a name on a warrant, falls entirely outside what *Carpenter* addressed. Every wrongful arrest driven by a facial

recognition match, every knock on a door justified by a risk score, occurs in a constitutional space the Court has not yet entered. This article enters it.

This article argues that when law enforcement uses an AI system to generate a person specific probabilistic inference and acts on that inference, it has conducted a search within the meaning of the Fourth Amendment, and that search requires a warrant or an established exception. The constitutional moment is not when the underlying data was collected. It is when the government relies on a person-specific output, an inference targeting a named individual, as the basis for action against that person. A raw dataset is inert. A list of location pings, purchase records, or facial images sitting in a database does not accuse anyone, flag anyone, or authorize action against anyone. An AI output attaching a risk score or identity match to a specific person is a different thing entirely. It is actionable state knowledge. The government can arrest on it, surveil on it, and prosecute on it. The person it targets is harmed by it. That harm, not the collection of data that preceded it, is where the Fourth Amendment's protection must attach.

This article proceeds in six parts. Part II explains how predictive AI surveillance systems operate, distinguishing place-based tools from person-based tools that generate individual risk scores and identity matches. Part III traces the Fourth Amendment's evolution from *Katz* through *Jones* to *Carpenter*, showing how each case moved doctrine closer to the problem AI inference presents without resolving it. Part IV develops the core argument: that the constitutional harm occurs when a person-specific inference is used as the basis for action, that AI inference is categorically different from human analysis, and that probabilistic outputs are structurally incompatible with the probable cause standard. Part V addresses the two most significant counter arguments. Part VI proposes a narrow, judicially administrable standard and considers whether that standard carries a practical remedy. Part VII concludes.

II. How Predictive AI Surveillance Works

Law enforcement agencies across the United States have integrated artificial intelligence into their investigative and patrol operations at a pace that has outrun the legal frameworks designed to govern them. This section does not attempt a comprehensive survey of AI surveillance technology. It establishes one factual premise on which the legal argument that follows depends: AI does not merely collect or organize data. It produces something new. That new thing, a risk score, a probability estimate, an identity match attached to a specific person's name, is what the government acts on, and it is what the law must reckon with.

Predictive AI surveillance tools fall into two broad categories. The first, place-based systems, analyze historical crime data to generate forecasts about where and when criminal activity is likely to occur. Platforms like PredPol, now rebranded as Geolitica, and ShotSpotter produce hotspot maps and acoustic gunshot alerts that police departments use to allocate patrol resources. These tools target locations and times, not individuals. The constitutional questions they raise, while real, are not the focus of this article.

The second category, person-based systems, is what this article addresses. These tools analyze data about specific individuals and produce outputs that target those individuals for police attention. The Chicago Police Department's Strategic Subject List, widely known as the Heat List, is the most documented example. The system assigned every person in its database a numeric risk score between zero and five hundred, calculated from factors including prior arrests, age, victimization history, and connections to others on the list. [3] A person with a high score could be visited at home by officers, placed in intervention programs, or subjected to intensified surveillance, all on the basis of a number generated by an algorithm. The score was not a finding that the person had done anything. It was a prediction about what they might do. By 2019, when the program was discontinued, the list contained approximately four hundred thousand names. [4] Facial.

recognition systems operate on a similar logic. These tools scan an image, typically from surveillance footage, and compare it against a database of photographs to generate a ranked list of potential matches. The system does not identify a suspect. It produces a probability: this image resembles this person in the database at this level of confidence. Detroit's Project Green Light program and similar systems in use across the country feed facial recognition outputs directly into investigative workflows. When a detective receives a match, what they receive is not evidence. It is an inference, a statistical judgment produced by a machine that the person in the surveillance image is the person in the database. In the cases of Robert Williams, Nijeer Parks, Michael Oliver, and Porcha Woodruff, that inference was wrong. In each case, police acted on it anyway. [5]

What unites these systems is the nature of their output. Before the algorithm runs, there is data: images, records, location pings, arrest histories. That data does not point at anyone. After the algorithm runs, there is something qualitatively different: a number attached to a name, a match attached to a face, a judgment that this specific person warrants police attention. That transformation, from raw data to person-specific inference, is the constitutional event this article argues the Fourth Amendment must address. The data that fed the algorithm may have been collected lawfully. The inference the algorithm produced is a different thing entirely, and it is the inference, not the data, that put Nijeer Parks in a jail cell for ten days. Understanding why that distinction matters constitutionally requires first tracing how Fourth Amendment doctrine has evolved to meet earlier generations of surveillance technology.

III. Fourth Amendment Doctrine and Its Digital Evolution

The Fourth Amendment provides that the right of the people to be secure in their persons, houses, papers, and effects against unreasonable searches and seizures shall not be violated, and that no warrants shall issue but upon probable cause. For most of American history, courts applied this guarantee to physical intrusions: an officer

entering a home, searching a car, or seizing papers. The digital age forced a reckoning. Three cases define the arc of that evolution. Each moved doctrine closer to the problem AI inference presents. None solved it.

A. *Katz v. United States*, 389 U.S. 347 (1967)

The modern Fourth Amendment framework begins with *Katz v. United States*, in which the Supreme Court held that the Amendment protects people, not places. Federal agents had attached a listening device to the outside of a public phone booth and recorded Charles Katz's end of a conversation without physically entering the booth. The Court held this constituted a search. Writing separately, Justice Harlan II articulated the test that has governed Fourth Amendment analysis ever since: a person enjoys constitutional protection when they have a subjective expectation of privacy that society is prepared to recognize as reasonable. [6]

Katz was a landmark, but its framework was built for a different kind of *surveillance* problem. The reasonable expectation of privacy test asks whether an individual had a reasonable expectation of privacy with respect to a specific act of government observation. It was designed for cases involving a defined moment of intrusion. It was not designed to address what happens when thousands of individually unremarkable data points, none of which would trigger constitutional protection on its own, are aggregated over months and synthesized by an algorithm into a probabilistic judgment about a specific person. *Katz* gives us the baseline. It does not give us the answer.

B. *United States v. Jones*, 565 U.S. 400 (2012)

United States v. Jones advanced the doctrine significantly, though the most important reasoning came not from the majority but from the concurrences. The majority held that attaching a GPS tracking device to Antoine Jones's vehicle and monitoring its movements for twenty-eight days constituted a trespassory search under the Fourth Amendment. [7] Five justices pressed the harder question in separate writings: what about surveillance that involves no physical intrusion at all? Justice Sotomayor wrote that GPS monitoring generates a precise,

comprehensive record of a person's movements that reflects a wealth of detail about her familial, political, professional, religious, and sexual associations. [8] Justice Alito, joined by three others, agreed that long-term surveillance of this kind was constitutionally different in kind from any single act of observation. [9]

Together these concurrences articulated what scholars have called the mosaic theory: that aggregating individually innocuous data points over time creates a constitutionally significant picture of a person's life, even if each individual point would not trigger Fourth Amendment protection standing alone. [10] When an AI system aggregates months of location data, purchase records, arrest histories, and social network connections to generate a risk score, it is doing exactly what the *Jones* concurrences warned about, and doing it faster, at greater scale, and with less human judgment than any detective could replicate. *Jones* brought doctrine to the edge of this problem. It did not cross it.

C. Carpenter v. United States, 585 U.S. 296 (2018)

Carpenter v. United States is the essential foundation for any Fourth Amendment argument about AI surveillance. The Court held that the government's acquisition of seven or more days of cell-site location information constitutes a search requiring a warrant, rejecting the argument that the third-party doctrine eliminated any constitutional protection for records voluntarily shared with a service provider. [11] Chief Justice Roberts acknowledged that seismic shifts in digital technology may require a different application of the Fourth Amendment's principles and located the constitutional interest not in the abstract fact that data existed somewhere but in the government's ability to chronicle a person's past movements and access what he called the privacies of life. [12]

But the *Carpenter* Court was emphatic about the limits of its holding. The opinion does not disturb conventional surveillance techniques and does not address other collection technologies. [13] Critically, it said nothing about what happens after collection, when an AI system takes that data and synthesizes it into a probabilistic inference targeting a

specific person. *Carpenter* addressed the moment data enters government hands. It did not address the moment an algorithm transforms that data into an actionable judgment about who you are, what you might do, and whether you should be the subject of law enforcement attention. Every person-based AI surveillance system in operation today produces outputs in that gap. The argument that follows explains why the Constitution must reach them.

IV. The Inference Is the Search: Why AI-Generated Outputs Constitute a Fourth Amendment Search

The preceding sections established two things: that person-based AI surveillance systems produce new, person-specific outputs that serve as the operative basis for law enforcement action, and that existing Fourth Amendment doctrine has moved progressively toward this problem without resolving it. This section makes the affirmative case. The argument proceeds in four steps, each building on the last.

A. The Constitutional Harm Occurs When the Inference Is Used, Not When the Data Is Collected

The first and most fundamental argument is that the Fourth Amendment's protection must attach when law enforcement relies on a person-specific AI output to act against a named individual, not at the earlier moment when the underlying data was collected. This argument follows directly from the logic of *Carpenter*, properly understood. It also requires being precise about where, within that later window, the constitutional line falls. It does not fall at the moment an algorithm generates an inference. It falls at the moment that inference becomes the operative basis for government action against a specific person. Generation without use is constitutionally inert in the same way collection without inference is inert: a database of matches no officer has acted on accuses no one. It is the act of relying on a person-specific output, to detain, arrest, surveil, or otherwise target an individual, that converts an algorithmic byproduct into a Fourth Amendment event.

A raw dataset is constitutionally inert. A database containing millions of facial images, location records, arrest histories, and social network

connections does not, by virtue of its existence, accuse anyone, flag anyone, or authorize action against anyone. The data points that fed the algorithm targeting Nijeer Parks had been collected and stored without constitutional incident. No Fourth Amendment violation occurred when that data was gathered. The violation occurred when the facial recognition system synthesized those inputs into a single output: this person is the suspect. That output was what the police acted on. That output was what put Parks in a cell.

This distinction between collection and use maps directly onto the analytical move the Supreme Court made in *Carpenter*. The Court located the constitutional interest not in the existence of data but in the government's capacity to transform that data into actionable knowledge about a specific person. [14] An AI inference is precisely that transformation, executed at a scale, speed, and level of comprehensiveness that no human analyst could replicate. If the transformation of cell-site data into a movement map required a warrant in *Carpenter*, the transformation of aggregated personal data into a person-specific risk score or identity match requires no less.

Paul Ohm has argued that *Carpenter* creates a rule of technological equivalence: if a technology gives police the power to gather information equivalent to a recognized Fourth Amendment search, using that technology is also a search. [15] The equivalence here is direct. The output of a facial recognition system is functionally equivalent to a detective pointing at a photograph and saying that is the person who committed this crime. The output of a risk-scoring algorithm is functionally equivalent to a detective saying this person is likely to commit a crime. Both outputs are actionable. Both authorize law enforcement intervention against a named individual. The Constitution does not permit the detective to make that judgment without probable cause. It should not permit the algorithm to do so either.

It is worth being explicit about what this argument does and does not claim, since the two can be conflated. It does not follow that every act of algorithmic inference is itself a search requiring prior judicial authorization. An AI system may generate outputs continuously, and the

government does not conduct a search each time an algorithm quietly returns a probability that no one acts on. What triggers the Fourth Amendment is the government's reliance on that output as the operative basis for action against a specific person, an arrest, a stop, a search, a knock at the door. This is not a retreat from the article's central claim; it is a specification of it. The inference is the search only once it is put to use. The title reflects the point at which the constitutional harm crystallizes, not a claim that generation and use are constitutionally identical moments requiring identical judicial process. Part VI develops this operative-basis principle into a standard courts can apply.

B. AI Inference Is Categorically Different From Human Analysis

The most intuitive objection to the argument advanced in Part A is that AI inference is simply a faster, more efficient version of what detectives have always done. A detective who examines a surveillance photograph and identifies a suspect has not conducted a new search. Neither, the argument goes, has a facial recognition system that does the same thing more quickly. This objection fails for three independent reasons.

The first reason is scale. A human detective reviewing evidence operates within the constraints of human cognition: limited working memory, finite time, and the practical inability to cross-reference more than a handful of data sources simultaneously. An AI system faces none of these constraints. The Chicago Strategic Subject List processed data on approximately four hundred thousand individuals simultaneously, cross-referencing arrest histories, victimization records, age, and social network connections to generate individualized risk scores. [16] No human analyst could replicate this process. When the Supreme Court held in *Carpenter* that the third party doctrine could not be mechanically applied to digital data because the nature of the intrusion was qualitatively different from anything the doctrine's authors contemplated, it was making exactly this point. [17]

The second reason is speed. Human analysis involves deliberation. A detective who reviews evidence, forms a hypothesis, and acts on it has engaged in a process that involves human judgment at each step, creating at least the possibility of pause and error correction before

action is taken. An AI system produces its output instantly and delivers it in a form that invites immediate action. There is no deliberative interval between the algorithm's conclusion and the officer's response. When Detroit police received a facial recognition match in the Robert Williams case, they acted on it the same day, without independent verification, and without any mechanism for Williams to challenge the basis for the identification before his arrest. [18]

The third reason is opacity. A human detective who forms a conclusion about a suspect can, at least in principle, explain the basis for that conclusion. That reasoning can be tested and subjected to adversarial scrutiny. The reasoning behind an AI output frequently cannot. The COMPAS recidivism algorithm used in *State v. Loomis* was proprietary. The defendant could not access the weights assigned to the factors that produced his risk score, could not challenge the data inputs, and could not test the model's validity as applied to his circumstances. [19] The Wisconsin Supreme Court upheld the use of the score at sentencing despite acknowledging these concerns. [20] The opacity problem is not incidental to AI surveillance systems. It is structural. When the government acts on a black box output to deprive a person of liberty, the Constitution's requirement that searches be reasonable demands more than the assurance that the box exists and produced a number.

Taken together, scale, speed, and opacity distinguish AI inference from human analysis in ways that make it more constitutionally dangerous, not equivalent. The *Carpenter* Court recognized that when a technology enables surveillance of a comprehensiveness that would have been impossible in an earlier era, the old rules do not automatically apply. AI inference is the technology for the next generation of Fourth Amendment cases.

C. Probabilistic Outputs Are Structurally Incompatible With Probable Cause

The third and most original argument is that AI-generated probabilistic outputs cannot satisfy the probable cause standard, and that acting on them as if they could represent a structural violation of the Fourth Amendment's core requirement of individualized suspicion.

Probable cause requires a fair probability, based on specific articulable facts, that a particular person has committed or will commit a particular crime. [21] The standard is individualized: it attaches to a specific person based on specific facts about that person's conduct or circumstances. It is not satisfied by a showing that a person belongs to a group that tends toward criminal activity, or that a person's characteristics correlate statistically with past offending. [22]

An AI risk score does not produce probable cause. It produces what Bernard Harcourt has called actuarial suspicion: a statistical judgment that a person resembles, in some algorithmically determined way, a population that has historically offended. [23] The COMPAS algorithm, ProPublica found, correctly predicted violent recidivism in only twenty percent of cases, meaning that for every person accurately flagged as likely to reoffend violently, four people were flagged who would not. [24] A fifty-three percent overall accuracy rate is only marginally better than a coin flip. [25] These are not the statistics of probable cause. They are the statistics of a guess dressed in algorithmic clothing.

This is not an objection to probability as such. Probable cause is itself a probabilistic standard, and courts already treat other probabilistic inputs, a trained narcotics dog's alert, an informant's tip, a drug-courier profile, as capable of contributing to it. What distinguishes those accepted inputs from an algorithmic risk score is not that they involve probability but how that probability is generated, tested, and corroborated. A canine's alert is anchored to a specific search of a specific vehicle or bag, and its reliability is tested against the dog's certification and field performance record. An informant's tip is weighed under the totality-of-the-circumstances test this Court adopted in *Gates* itself, which asks not whether the tip is probabilistic but whether the informant's basis of knowledge and veracity can be independently corroborated. [26] Both inputs are event-specific: they say something about this stop, this vehicle, this encounter. An algorithmic risk score is not. It says something about a population the subject statistically resembles, generated in advance of and independent from the specific encounter, and its reliability is typically unknown to the officer relying on

it and, as *Loomis* illustrates, sometimes unknown even to the court reviewing it. That is the distinction this article's probable cause argument turns on, not that the output is a probability, but that it is a categorical probability substituting for an individualized one, generated by a process that neither the officer nor the person it targets can test.

This distinction tracks, and extends, existing scholarship on predictive policing and reasonable suspicion. Andrew Ferguson has argued that predictive policing outputs function much like anonymous tips and should be subject to comparable corroboration requirements before they support a stop. [27] Elizabeth Joh has described the discretion embedded in algorithmic suspicion systems as a new and largely unregulated form of surveillance discretion. [28] Both bodies of work address reasonable suspicion, the lower threshold governing investigative stops, and both are principally concerned with place-based and area-based predictive tools. The move this article makes is twofold: from reasonable suspicion to probable cause, where the Fourth Amendment's particularity requirement is at its most demanding and has the least tolerance for a categorical substitute, and from area-based outputs to the person-based, name-attaching outputs, facial recognition matches and individual risk scores, that are this article's focus. An algorithmic output can inform probable cause the way a tip can, as one corroborated data point among several. It cannot substitute for probable cause the way this article's opening cases show it currently does.

The problem runs deeper than accuracy. Even a perfectly accurate probabilistic output would not satisfy probable cause, because probable cause is not a statistical concept. It does not ask whether a person is more likely than not to have committed a crime. It asks whether there are specific articulable facts about this person that justify the inference that they did. [29] A risk score answers a different question entirely: given what this person looks like statistically, how similar are they to people who have offended? The person is being evaluated as an instance of a category, not as an individual. The Fourth Amendment's particularity requirement is structurally incompatible with this mode of reasoning. [30]

This incompatibility is not merely theoretical. In the cases of Robert Williams, Nijeer Parks, Michael Oliver, and Porcha Woodruff, police obtained or executed arrest warrants on the basis of facial recognition matches without independent verification of the match's reliability. [31] In each case, the algorithmic output substituted for the individualized probable cause determination the Fourth Amendment requires. The result was not probable cause. It was actuarial suspicion translated into an arrest warrant, with no judicial check on the translation.

D. The Hallucination Problem: When AI Fabricates With Confidence

The structural incompatibility between AI outputs and constitutional standards is compounded by a well-documented failure mode unique to AI systems: hallucination. AI systems, particularly large language models and pattern-matching tools, do not merely make mistakes of degree. They generate outputs that are internally confident, authoritative in presentation, and factually wrong in ways that a human analyst would not be. The system does not flag its own uncertainty. It produces a result and presents it as a conclusion.

The legal system has already confronted this problem directly. In *Mata v. Avianca, Inc.*, attorneys submitted a brief to a federal court containing six case citations generated by ChatGPT. The cases did not exist. The AI had fabricated the case names, the reporters, the courts, and the holdings, and presented them in the standard format of real legal authority. [32] When confronted, counsel admitted they had not verified the citations against any legal database. The court sanctioned the attorneys for submitting AI-generated work product as reliable without independent verification, holding that the duty of candor to the tribunal and the obligation of professional competence require more than deference to an algorithm's output. [33]

The *Mata* case is not merely an anecdote about careless lawyers. It is a window into the specific danger AI outputs pose when treated as authoritative conclusions rather than starting points for human verification. A facial recognition system does not produce a match and then note its own error rate. A risk-scoring algorithm does not attach a

confidence interval that the reviewing officer is trained to interpret. The output arrives as a number or a name, and the institutional incentives of law enforcement, the pressure to clear cases, to act on available intelligence, to trust the technology procurement process that acquired the tool, all push toward treating that output as reliable. *Mata* illustrates what happens when that trust is misplaced in a context where the stakes are high, and the error is visible. In the wrongful arrest cases, the error was equally dramatic and the institutional pressures equally real, but the person who paid the price was not a lawyer facing sanctions. It was Nijeer Parks spending ten days in a cell.

The hallucination problem strengthens the constitutional case for a warrant requirement. A warrant does not merely require that police have evidence. It requires that a neutral magistrate evaluate that evidence before action is taken. In the context of AI-generated outputs, that requirement serves a function it has always served, but with heightened urgency: it inserts a human check between an algorithmically produced conclusion and the deprivation of a person's liberty. If the legal profession's response to AI hallucination is that lawyers must verify AI outputs before submitting them to courts, the Constitution's response to AI surveillance must be at least as demanding: that law enforcement must present AI outputs to a magistrate before acting on them against a named individual.

V. Counterarguments and Responses

No serious argument goes unanswered, and the argument advanced in this article faces two significant objections. The first challenges the premise that AI inference is a constitutionally distinct act. The second challenges the remedy. Both deserve honest engagement, and neither survives scrutiny.

A. *Lawful Collection Immunizes Downstream Analysis*

The most doctrinally grounded objection is that the Fourth Amendment violation, if any exists, occurred when the data was collected, not when it was analyzed. On this view, AI inference is simply investigative work

product: the government obtained data lawfully, a system processed it, and an officer drew a conclusion. A detective who reviews a lawfully obtained surveillance photograph and identifies a suspect has not conducted a new search. Neither, the argument goes, has a facial recognition system that performs the same function more efficiently.

This objection has surface plausibility, but it proves too much and is inconsistent with the reasoning of *Carpenter*. Taken to its logical conclusion, it would permit the government to compile every lawfully obtained data point about every person in the country into a single comprehensive profile and act on the resulting inferences without constitutional constraint, so long as each individual piece of data was collected lawfully. That cannot be the law. The *Carpenter* Court explicitly rejected the argument that lawful collection immunizes government use of the resulting data, holding that the comprehensiveness and intimacy of the information obtained, not the technical legality of its collection, determines whether a Fourth Amendment search has occurred. [34] The mosaic assembled from individually innocuous data points is qualitatively different from any single point, and the Constitution must reach the mosaic even when it cannot reach the points.

Applied to AI inference, the logic is identical and, if anything, stronger. The output of a person-based AI surveillance system is not merely a mosaic. It is a mosaic that has been interpreted, weighted, and translated into an actionable judgment about a specific individual. The raw data did not accuse Nijeer Parks. The algorithm did. The constitutional harm is located precisely where this article locates it: at the moment the inference is produced and acted upon.

Orin Kerr has argued that the sequential approach to Fourth Amendment analysis, evaluating each investigative step individually rather than treating the aggregate as a constitutional event, is essential to doctrinal coherence. [35] Kerr's concern is real: a rule that treats aggregation itself as constitutionally significant could sweep too broadly, capturing ordinary investigative practices that pose no serious

threat to privacy. [36] But this article's proposed standard does not adopt the mosaic theory in its broadest form. It does not argue that aggregation alone triggers Fourth Amendment protection. It argues that the production of a person-specific AI output that forms the operative basis for investigative action is the constitutional event. That is a narrower and more administrable rule than the mosaic theory, and it directly addresses Kerr's administrability concern without abandoning the insight that motivates it.

B. Warrant Requirements Would Cripple Law Enforcement

The second objection is practical rather than doctrinal. Requiring a warrant before law enforcement acts on an AI inference, the argument goes, would impose an impossible burden on agencies that rely on these tools for time-sensitive operations. Predictive policing systems generate thousands of outputs continuously. Facial recognition systems return matches in seconds. Requiring judicial approval before acting on each output would eliminate the operational utility of these technologies.

This objection mischaracterizes the rule this article proposes. The proposed standard does not require a warrant every time an AI system produces an output. It requires a warrant, or an established Fourth Amendment exception, before law enforcement takes person-specific investigative action based on that output. A police department may use a place-based predictive policing system to allocate patrol resources without constitutional constraint. A department may run a facial recognition search without a warrant, because running the search is not the constitutional event. The warrant requirement attaches at the moment the department acts on the person-specific result, using it as the operative basis to detain, arrest, or surveil a named individual.

Nor is this requirement operationally unworkable. The Fourth Amendment already requires probable cause and, in most circumstances, a warrant before police arrest a named individual. The rule this article proposes adds one requirement to that existing

framework: that the AI output on which the arrest is based cannot substitute for the probable cause determination. Police may still use AI tools. They may still act quickly in exigent circumstances under existing exceptions to the warrant requirement. What they may not do is treat an algorithmic output as the equivalent of probable cause and act on it without independent verification or judicial oversight. That is not a new burden. It is the existing constitutional framework applied to a new kind of evidence.

The Supreme Court rejected a nearly identical efficiency argument in *Riley v. California*, when it held that police must obtain a warrant before searching a cell phone incident to arrest, despite law enforcement's argument that immediate access to phone data was operationally essential. [37] The Court acknowledged the burden but held that the privacy interests at stake outweighed the efficiency concern. [38] The privacy interests implicated by person-based AI surveillance are at least as significant as those implicated by a cell phone search, and the potential for harm, as demonstrated by the wrongful arrests documented in this article, is correspondingly severe. Efficiency is a legitimate consideration in Fourth Amendment balancing. It is not a trump card.

VI. A Proposed Standard

Having established that person-based AI inference constitutes a Fourth Amendment search, and having answered the principal objections to that conclusion, this article turns to the practical question: what rule should govern law enforcement use of AI-generated outputs? The standard proposed here is intentionally narrow. It identifies the minimum that existing constitutional doctrine requires, states that minimum as a clear and administrable rule, explains why courts can adopt it without breaking new ground, and confronts directly whether the rule would carry any practical remedy once existing suppression doctrine is taken into account.

A. The Proposed Rule

When law enforcement uses an AI system to generate a person-specific

inference that forms the operative basis for investigative action against that individual, that inference constitutes a search within the meaning of the Fourth Amendment and requires either a warrant supported by probable cause or an established Fourth Amendment exception. Three elements of this rule require definition. The first is person-specific inference. This element distinguishes the outputs this rule covers from those it does not. A place-based predictive policing system that identifies a neighborhood or a time window as high-risk does not generate a person-specific inference. A hotspot map does not name anyone. A facial recognition match that attaches an identity to a named individual does. A risk score that assigns a number to a specific person in a database does. The rule applies when the AI output identifies and targets an individual, not when it identifies and targets a location or a population.

The second element is the operative basis for investigative action. This element limits the rule to cases where the AI output is the reason police act, not merely one consideration among many in a broader investigation. A detective who has developed probable cause through traditional investigative work and who uses a facial recognition match as one confirming data point among several is not acting on an AI inference as an operative basis. The rule attaches when the AI output is what moves the investigation from suspicion to action against a specific named individual, when it is, as it was in the Parks and Williams cases, the thing that generates the warrant.

The third element is that the rule uses existing Fourth Amendment framework rather than creating new doctrine. It does not require courts to adopt the mosaic theory, to abandon the third party doctrine entirely, or to impose technology-specific regulations that would require ongoing judicial management of evolving AI systems. It asks only that courts apply the warrant requirement and the probable cause standard, tools already in their possession, to a new category of evidence.

B. What the Standard Does Not Require

It is equally important to be precise about what this standard does not

require. The standard does not require that AI surveillance tools be accurate. Accuracy is an important policy concern, but it is not a Fourth Amendment category. A warrant issued on the basis of an AI output that has been subjected to proper judicial scrutiny and evaluated against independent corroborating evidence would satisfy this standard even if the underlying AI system has a meaningful error rate, provided the totality of the evidence established probable cause.

The standard does not require that AI tools be auditable or that their source code be disclosed. It requires judicial oversight of the decision to act on an AI output, not judicial inspection of the algorithm that produced it. And the standard does not prohibit the use of AI surveillance tools in law enforcement. It requires that before a person's liberty is constrained on the basis of an AI-generated output, the government satisfy the same constitutional standard it must satisfy before constraining liberty on any other basis: probable cause, evaluated by a neutral magistrate, with the specific individual identified with particularity. That is not a new requirement. It is the Fourth Amendment.

C. Why Courts Can Adopt This Standard

The proposed standard is judicially administrable because it requires no technological expertise to apply. A court asked to evaluate whether police acted on a person-specific AI inference as the operative basis for investigative action need not understand how the algorithm works, what data it used, or how it weighted its inputs. It need only ask two questions: did the AI output identify a specific individual, and did police use that output as the reason to act against that individual? If the answer to both questions is yes, the Fourth Amendment requires a warrant or an exception. If the answer to either question is no, the rule does not apply.

The standard is also consistent with the trajectory of Supreme Court doctrine. *Katz* established that the Fourth Amendment follows the person, not the place. *Jones* established that long-term aggregation of data creates constitutionally significant information that no single data

point would. *Carpenter* established that the comprehensiveness of digital surveillance can trigger Fourth Amendment protection regardless of the third-party doctrine. Each of these holdings reflects the same underlying principle: that the Constitution must reach new technologies when those technologies enable the government to intrude on the privacies of life in ways the Amendment was designed to prevent. [39] The proposed standard is the next application of that principle. It does not ask courts to change direction. It asks them to keep going in the direction they have already chosen.

D. Remedy: Why the Good-Faith Exception Does Not Swallow the Rule

A rule without a remedy is an invitation, not a requirement. The standard proposed above only has practical bite if violating it produces consequences, and the most immediate objection is that it will not. Under *United States v. Leon* and *Herring v. United States*, evidence obtained through a Fourth Amendment violation is not excluded where police acted in objectively reasonable reliance on a warrant, or in reasonable reliance on a system or record they did not know was flawed. [40] Wisconsin's highest court sidestepped a strikingly similar concern in *Loomis* by treating COMPAS's proprietary weighting as a limitation the sentencing court could work around through a written advisement of the tool's limitations, rather than a defect requiring exclusion of the score itself. [41] An officer who acts on a facial recognition match or a risk score without knowing its underlying error rate has a substantial argument for good-faith reliance, and if that argument prevails as a matter of course, the warrant requirement proposed here would rarely produce suppression, whatever its doctrinal force on paper.

This concern is real, but it does not defeat the proposed standard, for two reasons. First, the good-faith exception protects reasonable reliance on existing authorization. It does not protect the absence of authorization altogether. *Leon* excused police who executed a warrant a neutral magistrate had already approved. *Herring* excused a recordkeeping error inside an otherwise routine, lawfully authorized arrest. [42] Neither excuses proceeding without a warrant at all where

the Fourth Amendment required one. Once this article's standard is adopted, an arrest built on an unwarranted AI-driven inference is not a good-faith error committed inside an authorized search. It is the absence of the authorization the Fourth Amendment demands, and that absence is closer to the conduct Leon and Herring leave outside their protection than to the conduct those cases were designed to excuse.

Second, to the extent courts remain inclined to extend good-faith protection to officers who reasonably believed an AI output was accurate, that inclination is itself a reason to require the warrant step this article proposes, not a reason to abandon it. A magistrate's independent review before action is taken is the mechanism that keeps good-faith reliance from becoming a standing excuse: it moves the reasonableness inquiry from after the arrest, where the exclusionary rule operates and good faith can forgive an officer's error, to before the arrest, where a neutral judicial officer, not the arresting officer, bears the burden of evaluating the AI output's reliability in the first instance. A rule that front-loads that inquiry does more to prevent the underlying harm than a rule that leaves the inquiry to be litigated after the fact ever could. The proposed standard's remedy is prevention rather than suppression, and it is not defeated by a doctrine designed to excuse good-faith conduct within an authorized search when the point of the standard is to require the authorization in the first place.

VII. Conclusion

The Fourth Amendment was written for a world of physical intrusions: officers entering homes, seizing papers, searching persons. It has survived into the digital age because the Supreme Court has been willing, imperfectly and incrementally, to extend its principles to new technologies when those technologies enable the government to reach into the privacies of life in ways the Amendment was designed to prevent. *Katz* extended it to electronic eavesdropping. *Jones* brought it to GPS tracking. *Carpenter* applied it to the comprehensive digital record of a person's movements. Each extension was contested. Each was necessary. Each followed from the same underlying recognition that the

Constitution's guarantee of security against unreasonable government intrusion cannot be rendered meaningless by technological change.

AI-driven predictive surveillance is the next technology that demands this recognition. When a facial recognition system scans a database and attaches a name to a crime, it has done something the Fourth Amendment has always cared about: it has pointed the government at a specific person and authorized action against them. When a risk-scoring algorithm assigns a number to a named individual, and that number becomes the reason officers appear at their door, the same thing has happened. The government has made a judgment about a person, acted on it, and constrained their liberty. The fact that an algorithm made the judgment rather than a detective does not diminish its constitutional significance. It amplifies it. Algorithms operate at a scale no detective can match, at a speed that forecloses deliberation, through reasoning that neither the officer acting on the output nor the person targeted by it can examine or challenge, and with a documented capacity to fabricate conclusions with the same confident presentation as accurate ones.

Nijeer Parks spent ten days in jail because an algorithm was wrong and nothing in the constitutional framework required anyone to check. Robert Williams was arrested in front of his daughters because a facial recognition match substituted for the individualized probable cause determination the Fourth Amendment requires, and no court had yet said it could not. The lawyers in *Mata v. Avianca* faced sanctions because they submitted an AI's fabricated conclusions to a federal court without verification. In each case, the common thread is the same: an AI system produced an output, a human actor treated it as authoritative, and the consequences fell on someone else. The Constitution exists precisely to interrupt that chain before it ends in a cell.

The Supreme Court said in *Carpenter* that it would not allow the Constitution to be defeated by seismic shifts in digital technology. [43] AI-driven predictive surveillance is that shift, arrived and accelerating. The inference is the search. The warrant is required.

Notes

[1] Kashmir Hill, “Another Arrest, and Jail Time, Due to a Bad Facial Recognition Match,” *New York Times*, December 29, 2020, <https://www.nytimes.com/2020/12/29/technology/facial-recognition-misidentify-jail.html>.

[2] *Carpenter v. United States*, 585 U.S. 296 (2018).

[3] Andrew Guthrie Ferguson, *The Rise of Big Data Policing: Surveillance, Race, and the Future of Law Enforcement* 34–61 (New York University Press 2017).

[4] *Id.*

[5] American Civil Liberties Union, “More Than a Dozen Wrongful Arrests Due to Police Reliance on Facial Recognition Technology,” ACLU (2023), <https://www.aclu.org/news/privacy-technology/more-than-a-dozen-wrongful-arrests-due-to-police-reliance-on-facial-recognition-technology>.

[6] *Katz v. United States*, 389 U.S. 347, 360–61 (1967).

[7] *United States v. Jones*, 565 U.S. 400, 404–05 (2012).

[8] *Id.* at 415 (Sotomayor, J., concurring).

[9] *Id.* at 430 (Alito, J., concurring in the judgment).

[10] Orin S. Kerr, *The Mosaic Theory of the Fourth Amendment*, 111 *Mich. L. Rev.* 311, 313 (2012), <https://doi.org/10.36644/mlr.111.3.mosaic>.

[11] *Carpenter*, 585 U.S. at 310.

[12] *Id.* at 306, 310.

[13] *Id.* at 316.

[14] *Id.* at 310.

[15] Paul Ohm, *The Many Revolutions of Carpenter*, 32 *Harv. J.L. & Tech.* 357, 380 (2019).

- [16] Ferguson, *supra* note 3, at 34–61.
- [17] Carpenter, 585 U.S. at 310.
- [18] ACLU of Michigan, “*Williams v. City of Detroit*” (2021), <https://www.aclumich.org/cases/facial-recognition/>.
- [19] *State v. Loomis*, 881 N.W.2d 749, 767 (Wis. 2016).
- [20] *Id.* at 769.
- [21] *Illinois v. Gates*, 462 U.S. 213, 238 (1983).
- [22] *Ybarra v. Illinois*, 444 U.S. 85, 91 (1979).
- [23] Bernard E. Harcourt, *Against Prediction: Profiling, Policing, and Punishing in an Actuarial Age 4–5* (University of Chicago Press 2007).
- [24] Julia Angwin et al., “Machine Bias,” ProPublica (May 23, 2016), <https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>.
- [25] *Id.*
- [26] *Gates*, 462 U.S. at 230–31.
- [27] Andrew Guthrie Ferguson, *Predictive Policing and Reasonable Suspicion*, 62 Emory L.J. 259, 305 (2012).
- [28] Elizabeth E. Joh, *The New Surveillance Discretion: Automated Suspicion, Big Data, and Policing*, 10 Harv. L. & Pol’y Rev. 15 (2016).
- [29] *Gates*, 462 U.S. at 238.
- [30] *Groh v. Ramirez*, 540 U.S. 551, 557 (2004).
- [31] ACLU, *supra* note 5.
- [32] *Mata v. Avianca, Inc.*, 678 F. Supp. 3d 443, 448 (S.D.N.Y. 2023).
- [33] *Id.* at 461.
- [34] Carpenter, 585 U.S. at 310–12.

[35] Kerr, *supra* note 10, at 314–25.

[36] *Id.* at 350–68.

[37] *Riley v. California*, 573 U.S. 373 (2014).

[38] *Id.* at 403.

[39] *Carpenter*, 585 U.S. at 306.

[40] *United States v. Leon*, 468 U.S. 897 (1984); *Herring v. United States*, 555 U.S. 135 (2009).

[41] See *State v. Loomis*, 881 N.W.2d 749 (Wis. 2016) (permitting continued use of the score at sentencing so long as the presentence report includes a written advisement of the tool’s limitations, rather than requiring disclosure of its methodology or exclusion of the score itself).

[42] *Leon*, 468 U.S. 897; *Herring*, 555 U.S. 135.

[43] *Carpenter*, 585 U.S. at 304.

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